

ClipMind AI: Video Summarization & KeyMoments Detection

Milestone 4: Evidence Lens and AWS Deployment

1. My Work Responsibility

- **Evidence Lens:** a feature that shows the supporting transcript evidence and its timestamp together with each answer about a video.
- **AWS Deployment:** deploying the ClipMind AI website on AWS so it can be used and demonstrated online.

2. Project Objective

The objective of Milestone 4 was to add the Evidence Lens feature to ClipMind AI and to deploy the complete website on AWS so that it can be opened and demonstrated online.

3. Project Description and Main Features

The complete processing flow for this milestone is:

Upload, FFmpeg audio extraction, Whisper transcription with timestamped segments, Key Moments, Highlights, Keywords, Analytics, Evidence Lens, deployment on AWS.

Evidence Lens

- **Answer:** the response to the user's question about the video.
- **Evidence:** the exact transcript passage that supports the answer.
- **Timestamp:** the start and end time of that passage in the video.
- **Watch Evidence:** jumps the video player to the evidence moment so the answer can be checked against the video.

AWS Deployment

The Docker setup prepared in Milestone 2 was used to host the website on AWS.

- **Containerized stack:** a Docker Compose file runs three services: MongoDB, the FastAPI backend (built from the backend folder with its environment file), and a Caddy web server on ports 80 and 443.
- **Persistent storage:** named volumes keep the MongoDB data, uploaded videos, generated clips, and the Hugging Face and Whisper model caches, so the large AI models are not downloaded again every time the containers restart.

- **Deployment preparation:** the items carried over from Milestone 1, restricting CORS and MongoDB Atlas network access, were set for the deployed server instead of staying open as they were for local development.
- **Live website:** <https://clipmind.duckdns.org>

4. Challenges Faced

- **Learner role could not upload files:** uploading is controlled by role-based access control, so a learner account was rejected when it tried to upload a video. Uploads are done from a content creator or educator account, and learners use the processed results.
- **Large AI models on a cloud server:** Whisper and Hugging Face models take a lot of storage and time to download. Keeping them in named volumes avoids downloading them again after a restart or redeploy.

5. Conclusion

This milestone added Evidence Lens to ClipMind AI and moved the platform from local development to a live website on AWS. Answers now come with the supporting transcript evidence and timestamp, so users can check them directly against the video, and the whole flow from upload to analytics can be demonstrated online at <https://clipmind.duckdns.org>.